

Ques $f(x, y)$ = no. of different bits in their representation.
Return the sum of $f(a_i, a_j)$ for all $0 < i < j < n$ pairs in array a .

Approach

- We will check for 0 to 31^{st} bit as numbers are in integer range.
- For i^{th} bit we will count the number of elements which has their i^{th} bit set and unset, then
 $sum += curBitSum \cdot (setCount \cdot unsetCount)$
- We will do this for all 32 bits.
- As order matters but all pairs are 'just mixers' if we change order, $\therefore$ return $sum \cdot 2 \% (1e9+7)$

Codes

```
public static int solve(int[] arr) {  
    int MOD = (int) 1e9+7;  
    int sum = 0;  
    for (int i = 0; i < 32; i++) {  
        int setCount = 0, unsetCount = 0;  
        for (int j = 0; j < arr.length; j++) {  
            if ((arr[j] & (1 << i)) != 0) {  
                setCount++;  
            } else {  
                unsetCount++;  
            }  
        }  
    }
```

```
    sum = (((setCount % MOD) * (unsetCount % MOD)) % MOD  
            + sum) % MOD;  
}
```

```

    } return sum*2;
}

```

* Test Cases 1

input 1 + [1, 3, 5]

0 0 1

0 1 1

1 0 1

sum = 0

Dry Run! 0th bit + setCount = 3, unset = 0

$$\text{Sum} = \text{sum} + 3 \times 0 = \boxed{\text{sum} = 0}$$

1st bit + setCount = 1, unsetCount = 2

$$\therefore \text{sum} = 2$$

2nd bit + setCount = 1, unsetCount = 2

$$\therefore \text{sum} = 4$$

return sum*2 = 8

* Time Complexity

$$O(3i \times n) \Rightarrow O(n)$$

Space Complexity

$$O(1)$$

As we didn't use any additional space.